

Math toolkit for the delegation / coverage argument

Working notes

1 The two empty things

Think of a SPARQL result as a *table*.

Object	Notation	Meaning
Empty solution <i>set</i>	$\emptyset$	no rows: the pattern did not match
Empty solution <i>mapping</i>	$\mu_\emptyset$	one row, zero columns
Join identity	$\{\mu_\emptyset\}$	$\Omega \bowtie \{\mu_\emptyset\} = \Omega$

The cleanest instance is **ASK**, which is exactly what Δ computes:

$$\text{ASK } \{ \langle a \rangle \langle p \rangle \langle b \rangle \} = \begin{cases} \text{true} & \text{iff } \llbracket \cdot \rrbracket_G = \{\mu_\emptyset\}, \\ \text{false} & \text{iff } \llbracket \cdot \rrbracket_G = \emptyset. \end{cases}$$

So $\{\mu_\emptyset\}$ means *yes* and $\emptyset$ means *no*. Note $\{\mu_\emptyset\} \neq \emptyset$: it is a one-element set. Writing $\emptyset$ for “matched with no bindings” would invert the indicator. Notation: $\mu_\emptyset$ in Pérez–Arenas–Gutiérrez, μ_0 in SPARQL 1.1 §18.5, which also names $Z = \{\mu_0\}$ the join identity.

2 The setting as a parameter

Nothing in what follows fixes what $\mathcal{D}$ is. Every definition and both theorems use only three properties: sources hold graphs, query evaluation is monotone in the graph, and the reference answer is evaluation over the union of the sources. What varies between federation and traversal is not the algebra but *how the set of sources comes to be known*, so we make that a parameter.

Definition 1 (Discovery setting). A *discovery setting* is a tuple $W = \langle \mathcal{D}, \text{data}, \mathcal{D}_0, \text{disc} \rangle$: a set of sources $\mathcal{D}$, a map $\text{data} : \mathcal{D} \rightarrow 2^T$ giving the graph each one holds, a set of seeds $\mathcal{D}_0 \subseteq \mathcal{D}$, and a *discovery function* disc assigning to every BGP B a map $\text{disc}_B : \mathcal{D} \times 2^T \rightarrow 2^{\mathcal{D}}$, where $\text{disc}_B(d, \text{data}(d))$ are the sources that a retrieved d exposes while B is evaluated.

Definition 2 (Reachable set, reference answer). Iterating discovery from the seeds, let $\mathcal{D}_{i+1} := \mathcal{D}_i \cup \bigcup_{d \in \mathcal{D}_i} \text{disc}_B(d, \text{data}(d))$. The *reachable set* is $\text{reach}_B(W) := \bigcup_{i \geq 0} \mathcal{D}_i$, the *reference answer* to B is $\llbracket B \rrbracket_{G_{\text{reach}_B(W)}}$, and correctness of a delegation is always relative to it. Equivalently $\text{reach}_B(W)$ is the smallest set containing $\mathcal{D}_0$ and closed under disc_B , though nothing below uses that characterisation.

The query is a genuine parameter of discovery — c_{All} and c_{Match} differ in nothing else — so disc and reach carry it. A single B is fixed from here on and we drop the subscript, writing disc and $\text{reach}(W)$.

disc	$\mathcal{D}_0$	recovers
$\text{disc} \equiv \emptyset$	all members	classical federated query processing
all URIs occurring in $\text{data}(d)$	seed URIs	LTQP under c_{All}
URIs of triples matching some $tp \in B$	seed URIs	LTQP under c_{Match}
the subweb spec. inside $\text{data}(d)$	the user’s document	guided traversal
a link path expression from the query	seed URIs	LDQL

Definition 3 (State at time T , frontier, termination). A *state* at time T is a set $\mathcal{D}'(T)$ with $\mathcal{D}_0 \subseteq \mathcal{D}'(T) \subseteq \text{reach}(W)$, non-decreasing in T . Its *frontier* is

$$F(T) := \left(\bigcup_{d \in \mathcal{D}'(T)} \text{disc}(d, \text{data}(d)) \right) \setminus \mathcal{D}'(T),$$

the sources exposed by what has been retrieved but not yet retrieved themselves. *Discovery has terminated* at T if $F(T) = \emptyset$, equivalently $\mathcal{D}'(T) = \text{reach}(W)$.

Remark (Federation is the degenerate case). Classical federation is $\text{disc} \equiv \emptyset$ with $\mathcal{D}_0$ the configured member list. Then $\text{reach}(W) = \mathcal{D}_0$ and $F(0) = \emptyset$: discovery terminates before the first request. This is exactly why exclusive groups are sound there, and why the impossibility result below has nothing to say about it.

Remark (The reference answer is not the Web). $\llbracket B \rrbracket_{G_{\text{reach}(W)}}$ is evaluation over the union of the *reachable* sources, not over the Web. Taking disc to expose every URI and $\mathcal{D}$ to be every document would give full-web semantics, which is not finitely computable; the traversal instantiations keep $\text{reach}(W)$ bounded by the seeds and the criterion.

Remark (How Δ is obtained differs). In a federated setting Δ comes from cheap probes — ASK requests, a VoID description, an index — independently of retrieving the data. Under traversal there is no such probe: $\Delta(t, d)$ becomes known exactly when $\text{data}(d)$ does, so Δ collapses into the state. The source-description language is therefore strictly poorer in the traversal instantiations, which strengthens rather than weakens the impossibility argument.

Remark (Beyond RDF). Removing triple patterns and dereferencing leaves monotone conjunctive queries over horizontally partitioned data, where a fragment may be shipped whole to one partition exactly when no answer spans partitions. That is the classical localisation condition for distributed query processing; the RDF instance merely supplies particular notions of partition and of link.

3 Relevance and delegation

Notation. For a set of sources $A \subseteq \mathcal{D}$ we write $G_A := \bigcup_{d \in A} \text{data}(d)$ for the union of their graphs. The three instances used below are $G_{\mathcal{D}}$ (the whole federation), $G_{\mathcal{D} \setminus S}$ (the non-delegated sources) and G_A for an arbitrary cover A in the coverage section.

Definition 4 (Source relevance). Let $\mathcal{P}$ be the set of triple patterns and $\mathcal{D}$ the set of all sources. Write $\llbracket t \rrbracket_d$ for the evaluation of t over the data of d . The *relevance indicator* $\Delta : \mathcal{P} \times \mathcal{D} \rightarrow \{0, 1\}$ is

$$\Delta(t, d) = \begin{cases} 1, & \text{if } \llbracket t \rrbracket_d \neq \emptyset, \\ 0, & \text{otherwise,} \end{cases} \quad \mathcal{D}_t := \{d \in \mathcal{D} \mid \Delta(t, d) = 1\}.$$

Δ is obtained for the patterns of the query under evaluation only — one ASK probe per pattern per source, as in FedX.

Definition 5 (Federation instance). A *federation instance* is a function $\delta : \mathcal{D} \rightarrow 2^{\mathcal{T}}$ giving the graph held by each source. It is *consistent* with Δ for $t_1, \dots, t_n$, written $\delta \models \Delta$, if

$$\Delta(t_i, d) = 1 \iff \llbracket t_i \rrbracket_{\delta(d)} \neq \emptyset \quad \text{for all } i \in \{1, \dots, n\}, d \in \mathcal{D}.$$

Remark. Δ is what the engine *knows*; δ is what is *there*. Many δ give the same Δ , and the planner must be correct for all of them. Do not call this a “source assignment”: FedQPL Def. 8 already uses that term for a query plan.

Definition 6 (Delegation). Let B be a BGP and $B' \subseteq B$ the delegated fragment. Let $S \subseteq \mathcal{D}$ be a set of sources such that every $d \in S$ provides an interface $I_d = (L_d, \varrho_d)$ with $B' \in L_d$; that is, each delegated source accepts B' as a *single* request, returning $\varrho_d(B', \text{data}(d)) = \llbracket B' \rrbracket_{\text{data}(d)}$. The delegated plan obtains for B'

$$R := \left(\bigcup_{d \in S} \varrho_d(B', \text{data}(d)) \right) \cup \llbracket B' \rrbracket_{G_{\mathcal{D} \setminus S}},$$

and the delegation is *correct* if

$$R \bowtie \llbracket B \setminus B' \rrbracket_{G_{\mathcal{D}}} = \llbracket B \rrbracket_{G_{\mathcal{D}}}.$$

Remark (Where the request granularity lives). Semantically $\llbracket B' \rrbracket_{\text{data}(d)} = \bowtie_{t \in B'} \llbracket t \rrbracket_{\text{data}(d)}$, so over a single source one conjunctive request and $|B'|$ pattern requests return the same answers. The two *plans* are distinguished by the order of union and join:

$$\bigcup_{d \in S} \llbracket B' \rrbracket_{\text{data}(d)} \quad \text{versus} \quad \bowtie_{t \in B'} \bigcup_{d \in S} \llbracket t \rrbracket_{\text{data}(d)}.$$

Union outside the join is one request per source; join outside the union is one request per pattern per source. Only the first is delegation.

Remark (Capability and cost). The condition $B' \in L_d$ is what makes “single request” meaningful, and it is the point at which interface heterogeneity enters: a triple-pattern-only interface has $L_d = \mathcal{P}$ and admits no B' with $|B'| > 1$, while a composite resource advertising the path tp_1/tp_2 has $L_d = \{tp_1/tp_2\}$, so $B' \in L_d$ is exactly the requirement that the delegated fragment match what the resource publishes. Counting requests as in FedQPL’s sa-cost, delegation costs $|S|$ where the per-pattern route costs $|B'| \cdot |S|$.

Remark (Soundness is free). $\text{data}(d) \subseteq G_{\mathcal{D}}$ and $G_{\mathcal{D} \setminus S} \subseteq G_{\mathcal{D}}$, so $R \subseteq \llbracket B' \rrbracket_{G_{\mathcal{D}}}$; with $\bowtie$ monotone, the left-hand side is always a subset of the right. Delegation can never produce a wrong answer, only lose results. The whole content of the definition is completeness.

Remark (Sanity checks). $S = \emptyset$ gives $R = \llbracket B' \rrbracket_{G_{\mathcal{D}}}$: not delegating is trivially correct. $S = \mathcal{D}$ gives $R = \bigcup_{d \in \mathcal{D}} \llbracket B' \rrbracket_{\text{data}(d)}$: the no-inter-source-join condition. $S = \{k\}$ with k exclusive gives $R = \llbracket B' \rrbracket_{\text{data}(k)} = \llbracket B' \rrbracket_{G_{\mathcal{D}}}$: FedX’s exclusive group, and FedQPL Def. 7 with $\varphi = \text{req}_k^{B'}$.

Remark (What is lost). Since $\llbracket B \rrbracket_G = \llbracket B' \rrbracket_G \bowtie \llbracket B \setminus B' \rrbracket_G$, the gap between R and $\llbracket B' \rrbracket_{G_{\mathcal{D}}}$ consists exactly of the B' -solutions spanning a boundary introduced by the grouping: across the $S / \mathcal{D} \setminus S$ split, or between two distinct sources inside S . Such a solution is harmless when it has no join partner in $B \setminus B'$, since it would have been discarded anyway.

Remark (Why the two-block form does not work). The variant $\llbracket B \rrbracket_{\mathcal{D}} = \llbracket B \rrbracket_{\mathcal{D} \setminus \{d\}} \cup \llbracket B \rrbracket_d$ forbids only joins crossing the d boundary; it is blind to joins *inside* the block $\mathcal{D} \setminus \{d\}$. Take $\mathcal{D} = \{d, d_1, d_2\}$ and $B = \{t_1, t_2\}$ sharing a variable, with t_1 matching only at d_1 , t_2 only at d_2 , and the two joining. The equation holds, yet an inter-source join exists.

Theorem 1 (Correct delegation). *Let $B' = \{t_1, \dots, t_n\}$. Delegation of B' to S is correct for every federation instance consistent with Δ if and only if*

$$\left(\exists k \in S : \bigcup_{i=1}^n \mathcal{D}_{t_i} \subseteq \{k\} \right) \quad \text{or} \quad \left(\bigcup_{i=1}^n \mathcal{D}_{t_i} \right) \cap S = \emptyset.$$

Remark. All relevant sources lie inside a single delegated source, or none of them lies in S at all — the latter being the “delegating to an irrelevant source is harmless” case, which plain exclusivity does not capture. Anything in between admits a spanning witness. Both disjuncts are forced by the poverty of Δ : recording only per-pattern relevance, it cannot reveal whether a join actually crosses sources. The quantifier over instances is essential — on a single fixed instance the “only if” fails, since two sources holding identical data are both relevant yet delegating to either is correct.

4 Incomplete discovery

Definition 1 gives Δ over all of $\mathcal{D}$, so an engine holding Δ knows everything. To say anything about discovery we must separate what is true from what has been seen.

Definition 7 (Observation, completion, decision procedure). At time T the engine has retrieved $\mathcal{D}'(T)$ and knows $\text{data}(e)$ for each $e \in \mathcal{D}'(T)$; write $O(T) := (\mathcal{D}'(T), (\text{data}(e))_{e \in \mathcal{D}'(T)})$. It can therefore also compute the frontier $F(T)$, but knows nothing of $\text{data}(d)$ for $d \in F(T)$. A *completion* of $O(T)$ is any discovery setting $\widehat{W} = \langle \widehat{\mathcal{D}}, \widehat{\text{data}}, \mathcal{D}_0, \text{disc} \rangle$ agreeing with $O(T)$ on $\mathcal{D}'(T)$; the adversary's freedom is exactly the data behind the frontier. A *decision procedure* is a partial function Π from observations to pairs (B', S) with $S \subseteq \mathcal{D}'(T)$, and is *sound* if whenever $\Pi(O(T)) = (B', S)$, delegating B' to S is correct in *every* completion of $O(T)$.

Theorem 2 (No certificate under incomplete discovery). *Let $O(T)$ be an observation at which discovery has not terminated, i.e. $F(T) \neq \emptyset$. Then there are $B', S \subseteq \mathcal{D}'(T)$ and completions $\widehat{\mathcal{D}}_1, \widehat{\mathcal{D}}_2$ of $O(T)$ such that delegating B' to S is correct in $\widehat{\mathcal{D}}_1$ and incorrect in $\widehat{\mathcal{D}}_2$. Consequently no sound decision procedure can output (B', S) .*

Proof. Take $B' = B = \{t_1, t_2\}$ with $t_1 = (?x, p, ?y)$ and $t_2 = (?y, q, ?z)$, so that $B \setminus B' = \emptyset$ and the outer join in Definition 3 is with $\{\mu_\emptyset\}$, i.e. the identity. Let the observation be $\mathcal{D}'(T) = \{e\}$ with $\text{data}(e) = \{(a, p, b)\}$, and let $S = \{e\}$.

World 1. Let $\widehat{\mathcal{D}}_1 = \{e\}$. Here $\mathcal{D}_{t_1} = \{e\}$ and $\mathcal{D}_{t_2} = \emptyset$, so $\llbracket B' \rrbracket_{G_{\widehat{\mathcal{D}}_1}} = \emptyset$, and likewise $R = \emptyset$. The delegation is correct.

World 2. Let $\widehat{\mathcal{D}}_2 = \{e, d^*\}$ with $\text{data}(d^*) = \{(b, q, c)\}$, where $d^* \in F(T)$ sits on the frontier. Now

$$\llbracket B' \rrbracket_{G_{\widehat{\mathcal{D}}_2}} = \{ \{ ?x \mapsto a, ?y \mapsto b, ?z \mapsto c \} \} \neq \emptyset,$$

whereas both terms of R vanish:

$$\bigcup_{d \in S} \llbracket B' \rrbracket_{\text{data}(d)} = \llbracket B' \rrbracket_{\text{data}(e)} = \emptyset, \quad \llbracket B' \rrbracket_{G_{\widehat{\mathcal{D}}_2 \setminus S}} = \llbracket B' \rrbracket_{\text{data}(d^*)} = \emptyset,$$

since e has no t_2 -match and d^* has no t_1 -match. Hence $R = \emptyset$ and the answer is lost: the delegation is incorrect.

Both completions agree with $O(T)$ on $\mathcal{D}'(T)$, so no function of $O(T)$ can distinguish them. $\square$

Corollary 1 (Infinite web). *If $\text{reach}(W)$ is infinite then $F(T) \neq \emptyset$ at every time T , and hence no sound decision procedure can certify a delegation at any time.*

Proof. At any time T the engine has performed finitely many retrievals, so $\mathcal{D}'(T)$ is finite. Suppose $F(T) = \emptyset$. Then $\text{disc}(d, \text{data}(d)) \subseteq \mathcal{D}'(T)$ for every $d \in \mathcal{D}'(T)$, i.e. $\mathcal{D}'(T)$ is closed under discovery, and $\mathcal{D}_0 \subseteq \mathcal{D}'(T)$ by Definition 3. By induction on i we get $\mathcal{D}_i \subseteq \mathcal{D}'(T)$ for every i : the base case is the seed inclusion, and if $\mathcal{D}_i \subseteq \mathcal{D}'(T)$ then $\mathcal{D}_{i+1} = \mathcal{D}_i \cup \bigcup_{d \in \mathcal{D}_i} \text{disc}(d, \text{data}(d)) \subseteq \mathcal{D}'(T)$ by closure. Hence $\text{reach}(W) = \bigcup_i \mathcal{D}_i \subseteq \mathcal{D}'(T)$ is finite, contradicting the assumption. So $F(T) \neq \emptyset$, and Theorem 2 applies at every T . $\square$

Remark (Never, not merely hard). The corollary is stronger than a statement about cost. It is not that certification takes a long time on a large web; it is that at no finite time does a certificate exist. Infinite reachable sets are not exotic — a server generating a document per natural number, or per geographic area, suffices — so this is the regime the traversal instantiations actually live in. Contrapositively, certifying a delegation requires $\text{reach}(W)$ to be finite *and* to have been retrieved in full.

Remark (Why Δ alone cannot establish this). The witness must exhibit concrete triples. At the level of Δ the two worlds are distinguished only by $\Delta(t_2, d^*) = 1$, and that alone does not settle correctness: if d^* merely *duplicated* data already held by e , no new solution would appear and the delegation would remain correct. What breaks it is a triple that is new *and* joins across the boundary.

Remark (Relevance elsewhere is not by itself a problem). Discovering $\Delta(t, d) = 1$ for an undiscovered $d \notin S$ does *not* break the delegation. Since $d \in \mathcal{D} \setminus S$, the second term $\llbracket B' \rrbracket_{G_{\mathcal{D} \setminus S}}$ of R still collects everything d can answer on its own. Only solutions *spanning* S and its complement — or spanning two distinct sources inside S — are lost. This is the narrower fact the impossibility argument needs.

Remark (Side conditions). The construction needs B' connected (two patterns sharing a variable), $S \neq \emptyset$, and at least one pattern of B' matched within S ; otherwise there is nothing for the adversary's triple to join to. Taking $B' = B$ removes any further condition, since then no join partner in $B \setminus B'$ is required.

Remark (What this asks of the assumption). The argument turns entirely on the adversary being free to place arbitrary data behind an unfetched URI. Any fix must remove that freedom rather than improve the source descriptions: no refinement of Δ over $\mathcal{D}'(T)$ helps, because both worlds agree there. This is what coverage and authority declarations supply in the next section.

5 Coverage

Definition 8 (Coverage). $\text{cov} : \mathcal{D} \rightarrow 2^{\mathcal{D}}$, in two flavours:

$$d \in \text{cov}_{\text{agg}}(c) \iff \text{data}(d) \subseteq \text{data}(c), \quad d \in \text{cov}_{\text{idx}}(x) \iff x \text{ publishes } \Delta(\cdot, d).$$

An aggregator lets you *act*; an index lets you *decide*. Lift to sets by $\text{cov}(A) := \bigcup_{c \in A} \text{cov}(c)$, and call A a *cover* of $\mathcal{D}$ if $\text{cov}(A) = \mathcal{D}$.

Remark. cov is extensive, monotone and idempotent (coverage is transitive), hence a closure operator on $2^{\mathcal{D}}$. Covers form an up-set; the minimal ones are the objects of interest.

Lemma 1 (Answer equivalence). *If A is an aggregator-cover of $\mathcal{D}$ then $G_A = G_{\mathcal{D}}$, and therefore $\llbracket Q \rrbracket_{G_A} = \llbracket Q \rrbracket_{G_{\mathcal{D}}}$ for every monotone Q .*

Lemma 2 (Observation asymmetry). *Let $\mathcal{D}'_t(T) := \{d \in \mathcal{D}'(T) \mid \Delta(t, d) = 1\} \subseteq \mathcal{D}_t$. Then*

$$|\mathcal{D}'_t(T)| \geq 2 \implies |\mathcal{D}_t| \geq 2, \quad \text{but} \quad |\mathcal{D}'_t(T)| = 1 \not\Rightarrow |\mathcal{D}_t| = 1.$$

Non-exclusivity is soundly concludable from a finite observation; exclusivity is not. That asymmetry *is* the impossibility result.

Definition 9 (Discovery closure). Recall $\text{disc}(d, \text{data}(d))$ from Definition 1. A set A is *discovery-closed* if

$$\forall c \in A : \text{disc}(c, \text{data}(c)) \subseteq \text{cov}(A).$$

Lemma 3 (Frontier). *If A is discovery-closed and $\text{seeds} \subseteq A$, then $\text{reach}(\text{seeds}) \subseteq \text{cov}(A)$.*

This converts the unverifiable global condition “I have seen everything” into the local, checkable one “my frontier is covered”.

Theorem 3 (Decidability under coverage). *If at time T some $A \subseteq \mathcal{D}'(T)$ is discovery-closed with admissible index coverage, then $\mathcal{D}_t$ is computable from $\mathcal{D}'(T)$ alone, and exclusivity of k for $\{t_1, \dots, t_n\}$ is decidable at time T .*

Theorem 4 (Impossibility without coverage). *If $\text{cov}(\mathcal{D}'(T)) \subsetneq \mathcal{D}$, then for every k with $k \in \mathcal{D}_{t_i}$ for all i there is a completion of $\mathcal{D}$ consistent with every observation made up to T under which delegating $\{t_1, \dots, t_n\}$ to k is incorrect.*

Corollary 2. *Without coverage declarations, exclusivity is eventually computable but not finitely computable (Hartig & Freytag, Defs. 13–14). With an admissible discovery-closed index cover it becomes finitely computable.*

Remark. Choosing the cheapest $A \subseteq \mathcal{D}'(T)$ with $\text{cov}(A) = \mathcal{D}$ is weighted Set Cover: NP-hard, greedy gives $H_n \approx \ln n$. FedQPL already proves the analogous hardness for minimal source assignments by reduction from node cover (Def. 10 and Prop. 2), so that precedent is citable rather than new.

Remark. Coverage and the namespace scope claim are one mechanism: a claim over N covers exactly the sources whose data lies in S_N . Coverage is the extensional presentation, the scope claim the intensional one.

6 Authority claims: generic claim map and anchor mapping

Here $\mathbb{D}$ is the universe of sources, of which $\mathcal{D}$ above is the reachable part.

Definition 10 (Composite source). A source $c \in \mathbb{D}$ is *composite* when it holds no data of its own and is instead given by a set $\mathcal{D}_c \subseteq \mathbb{D}$ of sources it aggregates. It contains *all* of their data,

$$\text{data}(c) = \bigcup_{d \in \mathcal{D}_c} \text{data}(d),$$

and a request to c is evaluated over all of it. The aggregation relation is well-founded: no source aggregates itself, directly or through other composites. Sources that are not composite are *base* sources.

Remark (Degenerate composite). If $\mathcal{D}_c = \{d\}$ with d a base source, then $\text{data}(c) = \text{data}(d)$: as far as its data goes, c is d . If c moreover claims nothing, it is interchangeable with d under every definition below (same data, same admitted data, same answers). In proofs we may therefore treat a base source as a degenerate composite that claims nothing, and a single document that does claim as a degenerate composite that does.

Definition 11 (Claim, claim map). A composite source c may *declare* a claim, given by a selector $\sigma_c : 2^T \rightarrow 2^T$ that is *triple-wise*, i.e. $\sigma_c(G) = \{t \in G \mid \sigma_c(\{t\}) = \{t\}\}$. We say c *claims* a triple τ when $\sigma_c(\{\tau\}) = \{\tau\}$. The claim does not restrict what c returns: c still returns all of $\text{data}(c)$. It is a guarantee about the claimed triples only: they are to be taken from c and from no other source. Base sources cannot claim. Write $\mathcal{C} \subseteq \mathbb{D}$ for the composite sources that declare.

A *claim map* $\kappa : \mathbb{D} \rightarrow (2^T \rightarrow 2^T)$ assigns to every source a triple-wise selector κ_d , the triples it claims, with $\kappa_d \equiv (G \mapsto \emptyset)$ for every base source d . Taking declarations at face value gives the *naive* claim map

$$\kappa_d^0 = \begin{cases} \sigma_d, & \text{if } d \in \mathcal{C}, \\ G \mapsto \emptyset, & \text{otherwise.} \end{cases}$$

Given a claim map κ , the *admitted data* of a source is

$$\theta_\kappa(d, G) = \{t \in G \mid t \in \kappa_d(G) \text{ or } t \notin \bigcup_{c \in \mathbb{D}} \kappa_c(G)\}. \quad (1)$$

A claimed triple is admitted only from a source that claims it; an unclaimed triple is admitted from wherever it occurs. Evaluation is carried out over admitted data rather than stored data: for $A \subseteq \mathbb{D}$ we replace G_A of Definition 6 by

$$G_A^\kappa = \bigcup_{d \in A} \theta_\kappa(d, \text{data}(d)).$$

Remark (Base sources guarantee nothing). A base source d has $\kappa_d \equiv \emptyset$, so $\theta_\kappa(d, \text{data}(d))$ is just its data minus whatever some composite claims. It contributes data but never a guarantee about it. A composite contributes everything it contains except triples claimed by *other* composites, and guarantees only its claim. No data is lost when a claim is withdrawn: the triples simply become unclaimed and are admitted everywhere.

Nothing so far prevents a source from claiming data it has no business speaking for: under κ^0 any composite can claim, say, all of DBpedia and thereby silence every other source on it.

Definition 12 (Anchor mapping). An *anchor mapping* ρ assigns to every source d a triple-wise selector ρ_d , its *anchor*: the triples d is entitled to claim. The anchor is part of the setting, not of any declaration: a source cannot choose its own. The *anchored* claim map is

$$\alpha_d(G) = \kappa_d^0(G) \cap \rho_d(G),$$

so a source claims what it both declares and is entitled to. From here on we evaluate over G_A^α , with θ_α given by (1).

α is a claim map in the sense of Definition 11: an intersection of triple-wise selectors is triple-wise, and a base source has $\kappa_d^0 \equiv \emptyset$, hence $\alpha_d \equiv \emptyset$. It only withdraws claims, never adds them, since $\alpha_d \subseteq \kappa_d^0$ pointwise. No case split is needed: a source that does not declare claims nothing, and an illegitimate claim shrinks to its legitimate part rather than being rejected whole. A composite claiming both DBpedia and its own namespace keeps the latter.

Remark (Namespace instantiation). On the Web, take $\text{ns}(c)$ the namespace of the IRI of c and

$$\rho_c(G) = \{ t \in G \mid \text{subj}(t) \in \text{ns}(c) \}.$$

ρ_c depends only on the identity of c , so the engine knows it as soon as it sees c , whatever remains behind the frontier. The anchor constrains what c may *claim*, not what it may *aggregate*; a requirement that $\mathcal{D}_c$ consist of documents in $\text{ns}(c)$ belongs to Definition 10 or to the instantiation, not here.

Assumption 1 (Disjoint anchors). *Distinct composites have disjoint anchors: for $c \neq c'$, $\rho_c(G) \cap \rho_{c'}(G) = \emptyset$ for every G .*

As $\alpha_c \subseteq \rho_c$, it follows that no triple is claimed under α by two distinct sources, which is the form in which the assumption is used below. It is a property of the world rather than of what sources declare: for namespaces it says that no composite's namespace extends another's, since prefixes neither of which extends the other select disjoint sets of subjects. Relaxing it is the overlapping-sources case.

Lemma 4 (Claimed triples occur only in their claimant). *Under Assumption 1, let c claim τ under α . Then for every source $d \neq c$, $\tau \notin \theta_\alpha(d, \text{data}(d))$. Consequently, for every $A \subseteq \mathbb{D}$, $\tau \in G_A^\alpha$ implies $c \in A$ and $\tau \in \text{data}(c)$.*

Proof. Suppose $\tau \in \theta_\alpha(d, \text{data}(d))$ for some $d \neq c$. Then $\tau \in \text{data}(d)$, and as α_c is triple-wise, $\tau \in \alpha_c(\text{data}(d)) \subseteq \bigcup_{c'} \alpha_{c'}(\text{data}(d))$. So the second clause of (1) fails and the first must hold: d claims τ , contradicting Assumption 1. Hence $\tau \in G_A^\alpha$ can only be admitted from c , which requires $c \in A$ and $\tau \in \text{data}(c)$. $\square$

Definition 13 (Pattern within a claim). A triple pattern t is *within the claim* of a composite c if c claims every triple matching t : $\alpha_c(\{\mu(t)\}) = \{\mu(t)\}$ for every μ with $\text{dom}(\mu) = \text{var}(t)$. A BGP is within the claim of c if each of its patterns is.

Theorem 5 (Delegation to a claim). *Under Assumption 1, let $B' \subseteq B$ be non-empty and within the claim of a composite $c \in \mathcal{D}(T)$, and let $\mathcal{D} = \text{reach}(W)$. Then delegating B' to $S = \{c\}$ is correct with respect to $G_{\mathcal{D}}^\alpha$, and this holds at every time T , whether or not $F(T) = \emptyset$.*

Proof. Write $G_d := \theta_\alpha(d, \text{data}(d))$ and $G := G_{\mathcal{D}}^\alpha$. Every triple matching a pattern of B' is claimed by c , so by Lemma 4:

- (i) $\llbracket t \rrbracket_{G_d} = \emptyset$ for every $t \in B'$ and every $d \neq c$;

- (ii) every triple of $\text{data}(c)$ matching some $t \in B'$ is claimed by c and by no one else, hence lies in G_c by the first clause of (1); with $G_c \subseteq \text{data}(c)$ this gives $\llbracket B' \rrbracket_{\text{data}(c)} = \llbracket B' \rrbracket_{G_c}$.

For a BGP, $\mu \in \llbracket B' \rrbracket_G$ iff $\mu(t) \in G$ for every $t \in B'$; by (i) each such $\mu(t)$ lies in G_c , so $\llbracket B' \rrbracket_G = \llbracket B' \rrbracket_{G_c} = \llbracket B' \rrbracket_{\text{data}(c)}$ by (ii). The second term of R is $\llbracket B' \rrbracket_{G_{\mathcal{D} \setminus \{c\}}} = \emptyset$ by (i), as $B' \neq \emptyset$. Hence $R = \llbracket B' \rrbracket_{\text{data}(c)} = \llbracket B' \rrbracket_G$, and

$$R \bowtie \llbracket B \setminus B' \rrbracket_G = \llbracket B' \rrbracket_G \bowtie \llbracket B \setminus B' \rrbracket_G = \llbracket B \rrbracket_G.$$

Nothing in the argument refers to $\mathcal{D}(T)$ beyond $c \in \mathcal{D}(T)$: (i) holds for every $d \neq c$, discovered or behind the frontier. $\square$

Remark (Why this escapes Theorem 2). The adversary of Theorem 2 wins by placing a triple behind the frontier that joins across the delegation boundary. If that triple matches a pattern of B' it is claimed by c , so it is not admitted anywhere but at c and cannot join. The claim removes exactly the adversary's freedom over the data behind the frontier.

Remark (We never delegate to base sources). Definition 13 requires a claim, and base sources have $\kappa_d \equiv \emptyset$, so no non-empty BGP is within the claim of a base source. By Remark 6 a single document can still be a delegation target, but only by being a degenerate composite that declares a claim.

Remark (What is still open). Definition 13 is strong: for subject-namespace claims a pattern is within the claim only if its subject is a constant in the namespace. Patterns with a variable subject need the anchor-variable argument, filtering c 's answer to solutions whose matched triples c claims.